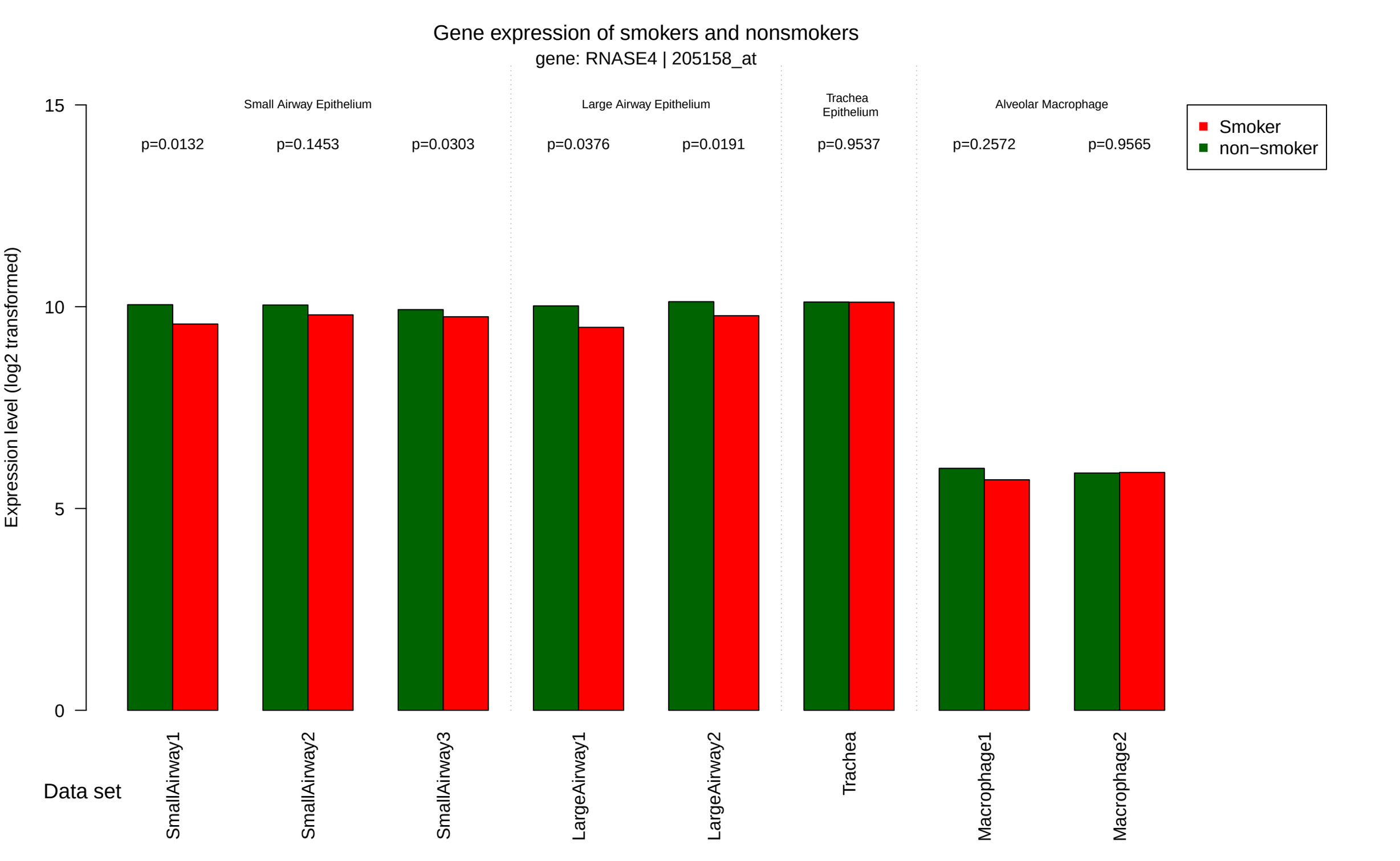
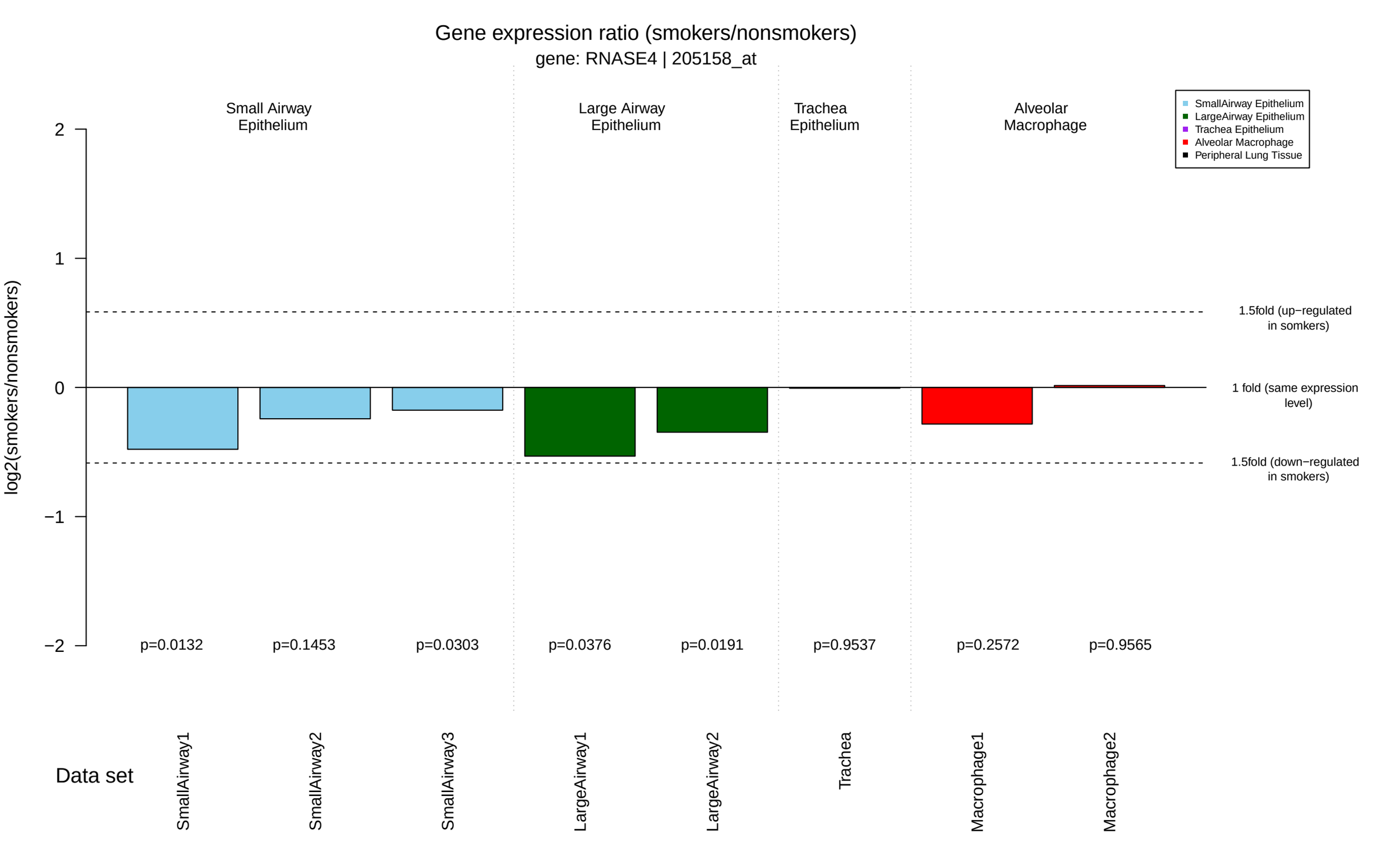
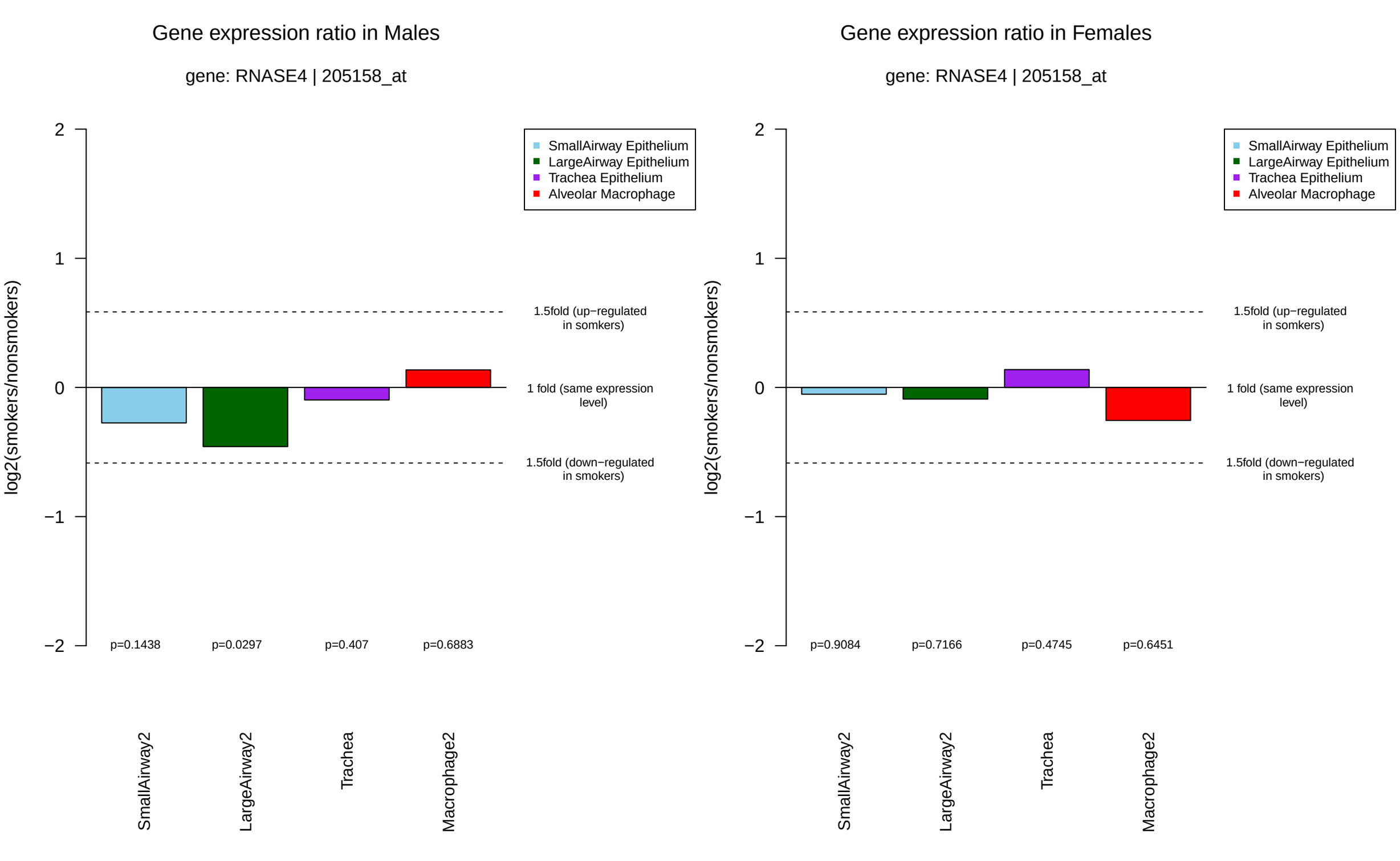
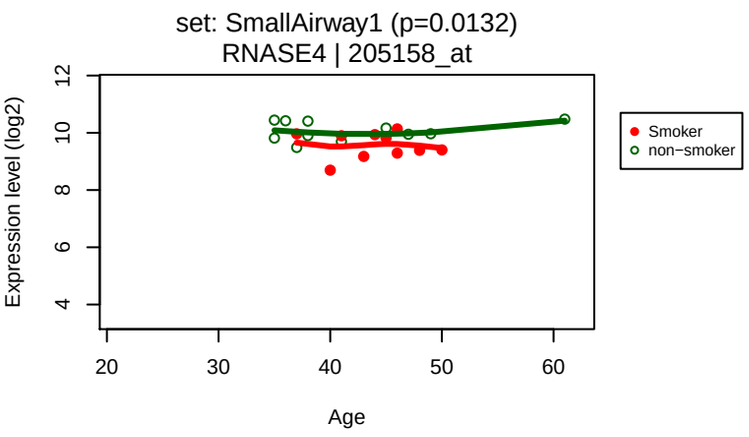
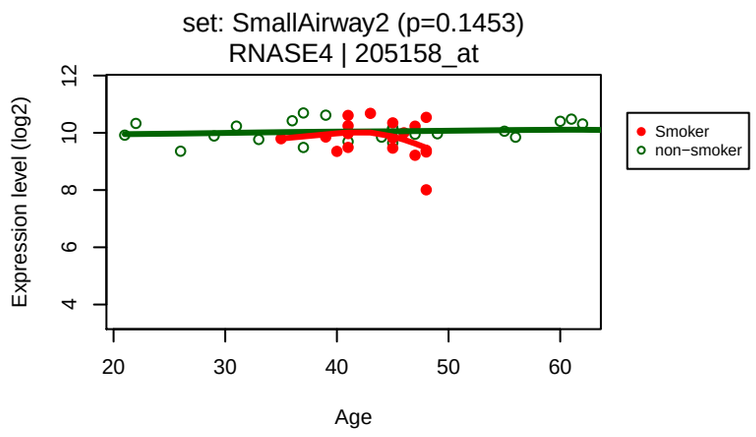
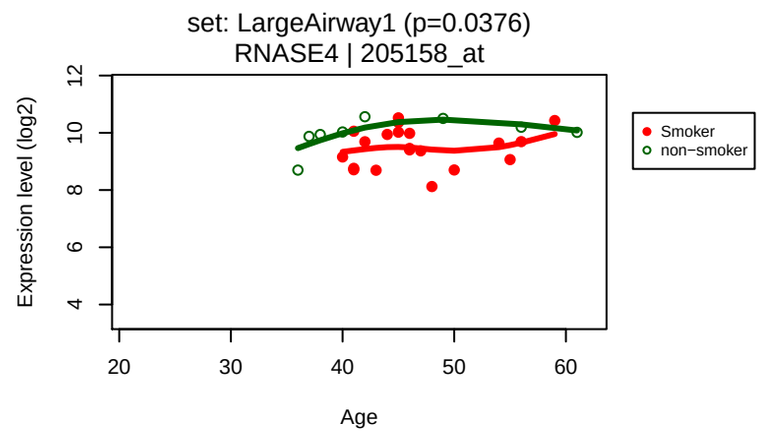

Gene expression of smokers and nonsmokers
gene: RNASE4 | 205158_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0132
p=0.1453
p=0.0303
p=0.0376
p=0.0191
p=0.9537
p=0.2572
p=0.9565
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

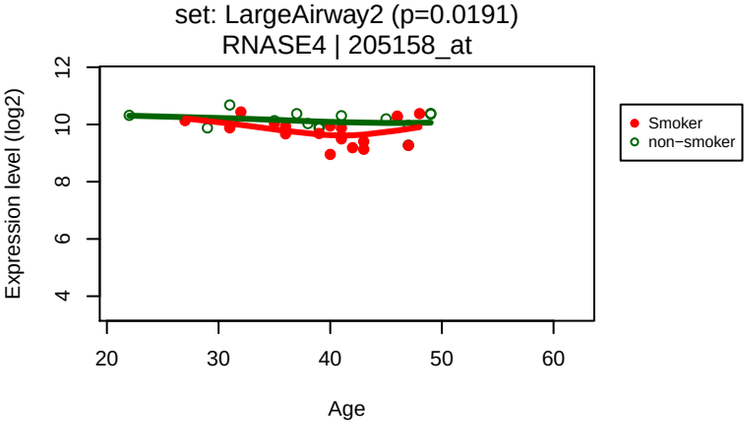

Gene expression ratio (smokers/nonsmokers)
gene: RNASE4 | 205158_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0132
p=0.1453
p=0.0303
p=0.0376
p=0.0191
p=0.9537
p=0.2572
p=0.9565
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

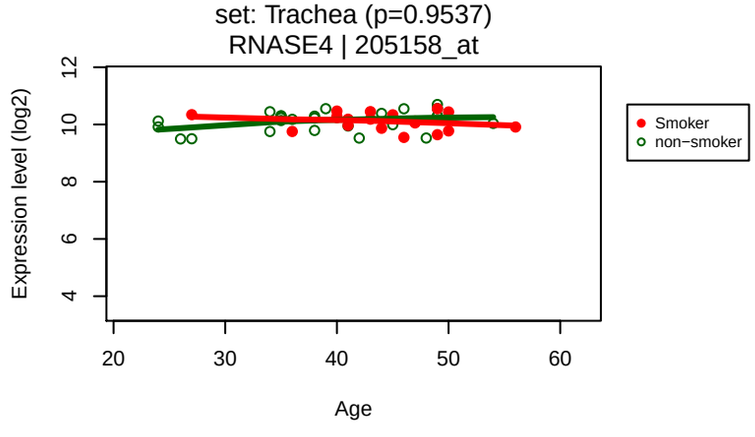

Gene expression ratio in Males
gene: RNASE4 | 205158_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1438
p=0.0297
p=0.407
p=0.6883
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNASE4 | 205158_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9084
p=0.7166
p=0.4745
p=0.6451
SmallAirway2
LargeAirway2
Trachea
Macrophage2

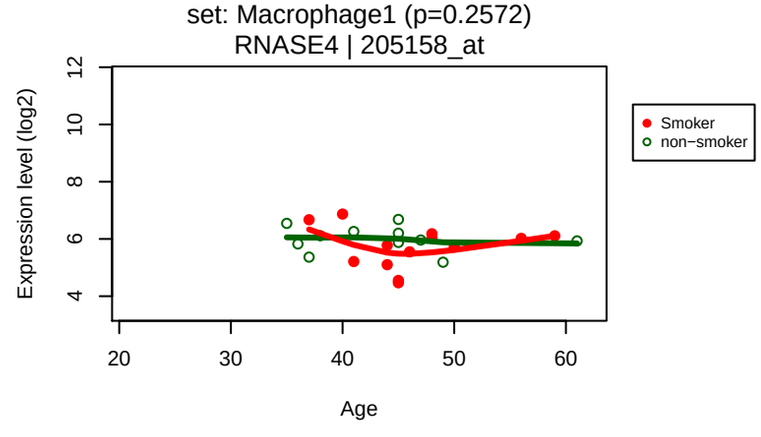

set: SmallAirway1 (p=0.0132)
RNASE4 | 205158_at
Expression level (log2)
Age
Smoker
non−smoker

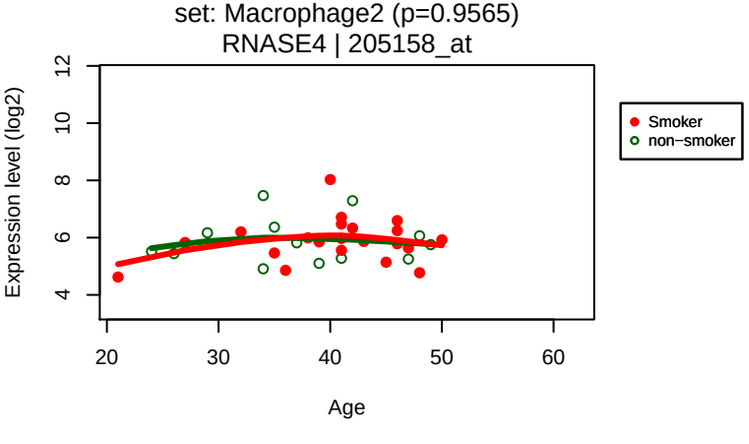

set: SmallAirway2 (p=0.1453)
RNASE4 | 205158_at
Expression level (log2)
Age
Smoker
non−smoker

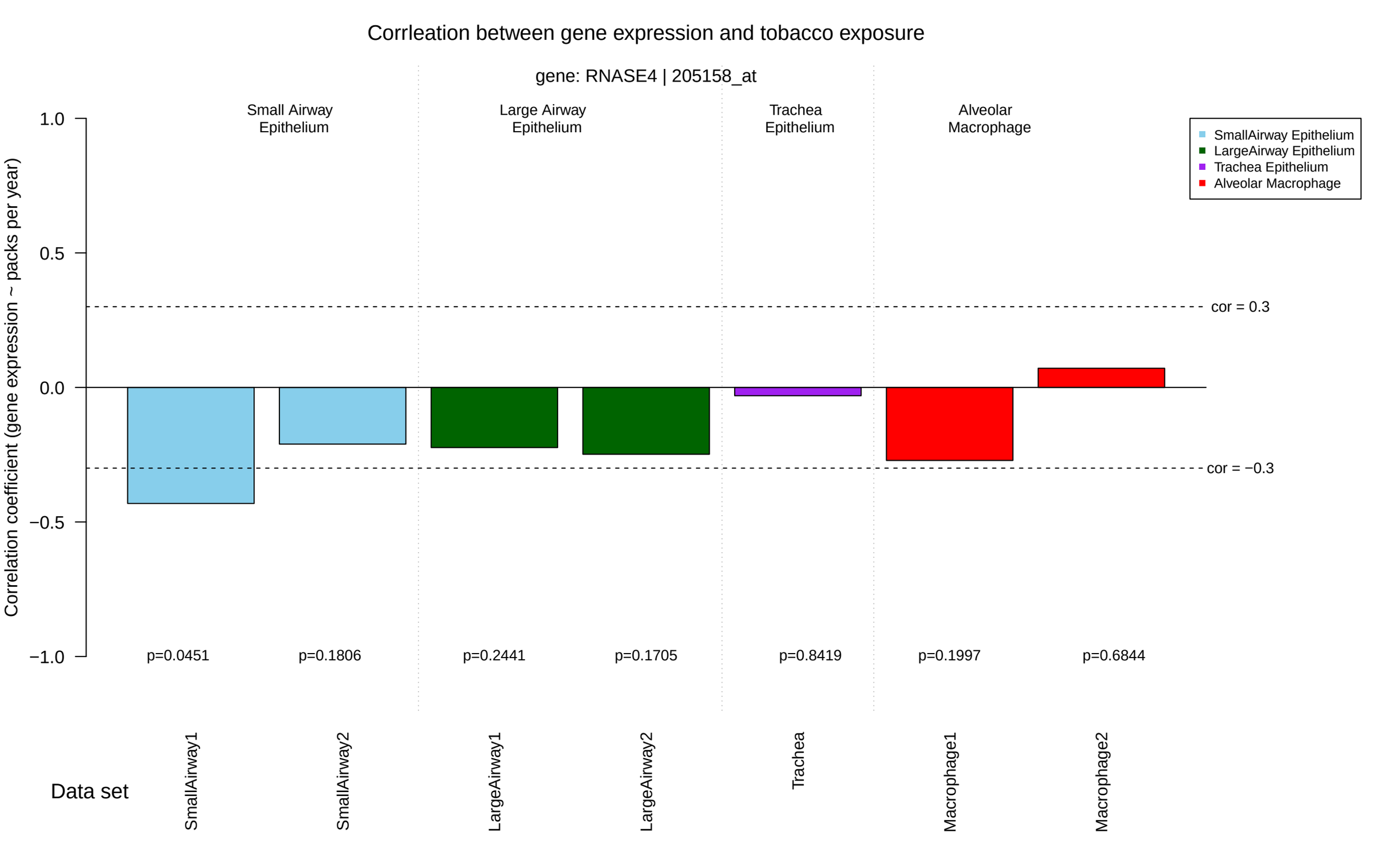

set: LargeAirway1 (p=0.0376)
RNASE4 | 205158_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0191)
RNASE4 | 205158_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9537)
RNASE4 | 205158_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2572)
RNASE4 | 205158_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9565)
RNASE4 | 205158_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNASE4 | 205158_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0451
p=0.1806
p=0.2441
p=0.1705
p=0.8419
p=0.1997
p=0.6844
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2